CYBR 4361 Project: NIST CSF Risk & Governance Assessment of a UT Dallas Student Organization

Project Summary: In this project, your team will act as an external cybersecurity risk advisory squad evaluating a real campus student organization (e.g., academic club, student society, sports club, or campus media group). You will use the NIST Cybersecurity Framework (CSF) 2.0 as a practical, interactive risk assessment tool. Using non-technical observation and a brief operational interview, you will evaluate the organization's governance, data handling, and access controls, auto-calculate its security maturity score, and present a prioritized 30-day remediation roadmap.

The 10 Curated NIST CSF 2.0 Subcategories: Evaluate your target student organization across these 10 core subcategories covering all six NIST CSF 2.0 functions: https://www.nist.gov/cyberframework

Function	**Subcategory ID**	**Subcategory Description**	**Operational Focus / Interview Questions**
Govern	GV.OC-01	Organizational Context	Does the club understand university data privacy rules when collecting member info or dues?
Govern	GV.RM-01	Roles & Responsibilities	Is there a designated tech/data officer, or does whoever is president handle all accounts and files?
Identify	ID.AM-01	Asset Management	Does the club maintain a list of all official assets (website domains, POS readers, social accounts)?
Identify	ID.RA-01	Risk Assessment	Has the executive board identified what happens if their primary email, Discord, or Instagram is hacked?
Protect	PR.AA-01	Access Management	Do officers use personal accounts and shared plain-text passwords, or do they enforce MFA?
Protect	PR.DS-01	Data Security	Are member rosters and financial records stored on personal drives or official university cloud storage?
Detect	DE.CM-01	Continuous Monitoring	Does the club routinely audit who currently has admin access to their group chat, drive, or social media?
Respond	RS.MA-01	Incident Management	Is there a clear procedure if an officer's account is compromised or financial funds are diverted?
Respond	RS.CO-02	Stakeholder Notification	Who is responsible for notifying university student affairs or members if sensitive data is exposed?
Recover	RC.RP-01	Recovery / Offboarding	When an officer graduates or leaves, is there a checklist to revoke access and change shared passwords?

Project Schedule

Phase 1: Target Outreach & Setup. Select a campus student organization. Reach out to a club officer (President, Vice President, Treasurer, or Tech Chair) using this sample script:

"Hi! We are undergraduate students in the Cybersecurity Governance & Risk Management course. For our course project, we are building a high-level NIST framework risk assessment for student orgs. May we conduct a brief 10-minute operational interview with an officer to help us build a free, confidential cybersecurity recommendation deck for your club?"

Phase 2: Data Collection & Audit

- Conduct your 10-minute interview with the student org officer.
- Perform your passive observational check (inspect public web forms, sign-up links, and public documents).
- Document your detailed justification/evidence for each of the 10 controls.

Phase 3: Excel Tool Build. Build an interactive 4-tab Excel workbook (.xlsx or Google Sheets):

1. Target Scope Tab: Organization name, mission, officer interviewed, and operational summary (e.g., *"Student club with 45 active members, collects $20 annual dues via Venmo, maintains an Instagram page, and uses a shared Google Drive"*).
2. Assessment Tab:
    - List the 10 NIST CSF controls.
    - Add a Data Validation Dropdown for Maturity Ratings:
        - 0 - Not Implemented (Ad hoc / Non-existent)
        - 1 - Partially Implemented (Informal / Inconsistent)
        - 2 - Largely Implemented (Documented / Standardized)
        - 3 - Fully Implemented (Optimized / Automated)
    - Include a mandatory "Evidence & Justification" text column for each rating.
3. Scoring Logic Tab: Use basic formulas (AVERAGEIF, SUM) to roll up ratings into function scores (Govern, Identify, Protect, Detect, Respond, Recover) and an overall security maturity percentage.
4. Executive Dashboard Tab: Apply Conditional Formatting (Red = $0.0 - 1.2$, Yellow = $1.3 - 2.2$, Green = $2.3 - 3.0$) to create a visual maturity heatmap and highlight the top 3 critical gap areas.

Phase 4: Presentation & Submission. Draft an executive presentation deck to deliver in class along with a demonstration of your Streamlit dashboard:

- Slide 1: Title & Target Student Org Profile
- Slide 2: Current Operational Environment & Data Handled (rosters, financial dues, social accounts)

- Slide 3: NIST CSF Assessment Breakdown (Showcasing your Streamlit app)
- Slide 4: Top 3 Operational Risk Exposures Identified (e.g., *Offboarding gap, unencrypted payment data on personal drives*)
- Slide 5: Prioritized Remediation Roadmap (Actionable, zero-cost fixes for the student org, such as adopting university cloud storage or setting up an offboarding checklist)

Step-by-Step Instructions: Launching Your Dashboard

To convert your completed Excel assessment into an interactive web dashboard using Streamlit Community Cloud, follow these 4 non-technical steps:

Step 1: Download the Starter Script from Canvas

1. In Canvas, navigate to the Resources tab in Module: Project.
2. Download app.py (the pre-configured Streamlit Python script) and requirements.txt.
3. Save both files in a folder on your computer alongside your team's completed Excel workbook. *(Ensure your Excel file is named nist_assessment.xlsx)*.

Step 2: Upload Files to a Personal GitHub Repository

1. Log in to GitHub (create a free account if you do not have one).
2. Click the **"+"** icon in the top right corner → Select **New repository**.
3. Name your repository (e.g., nist-csf-dashboard) and set visibility to **Public**. Click **Create repository**.
4. On the main repository page, click **Add file → Upload files**.
5. Drag and drop your three files into the browser window:
    - app.py
    - requirements.txt
    - Your completed nist_assessment.xlsx sheet
6. Click the green **Commit changes** button at the bottom.

Step 3: Deploy Free on Streamlit Cloud

1. Navigate to **share.streamlit.io** and log in using your GitHub account.
2. Click **Create app** (or **New app**).
3. Select your GitHub repository (nist-csf-dashboard), keep the branch set to main, and confirm app.py is selected as the main file.

4. Click **Deploy.** Within 60 seconds, Streamlit will compile your web application and generate a public web URL (e.g., https://yourname-nist-csf-dashboard.streamlit.app).

Step 4: Submission & Presentation

- Paste your live Streamlit web link into your presentation deck.
- During your final group presentation, open your live URL on the projector to filter NIST functions, display metrics, and walk through your risk heatmap live in front of the class.

Here is the list of interactive features and visualizations included in the app.py file:

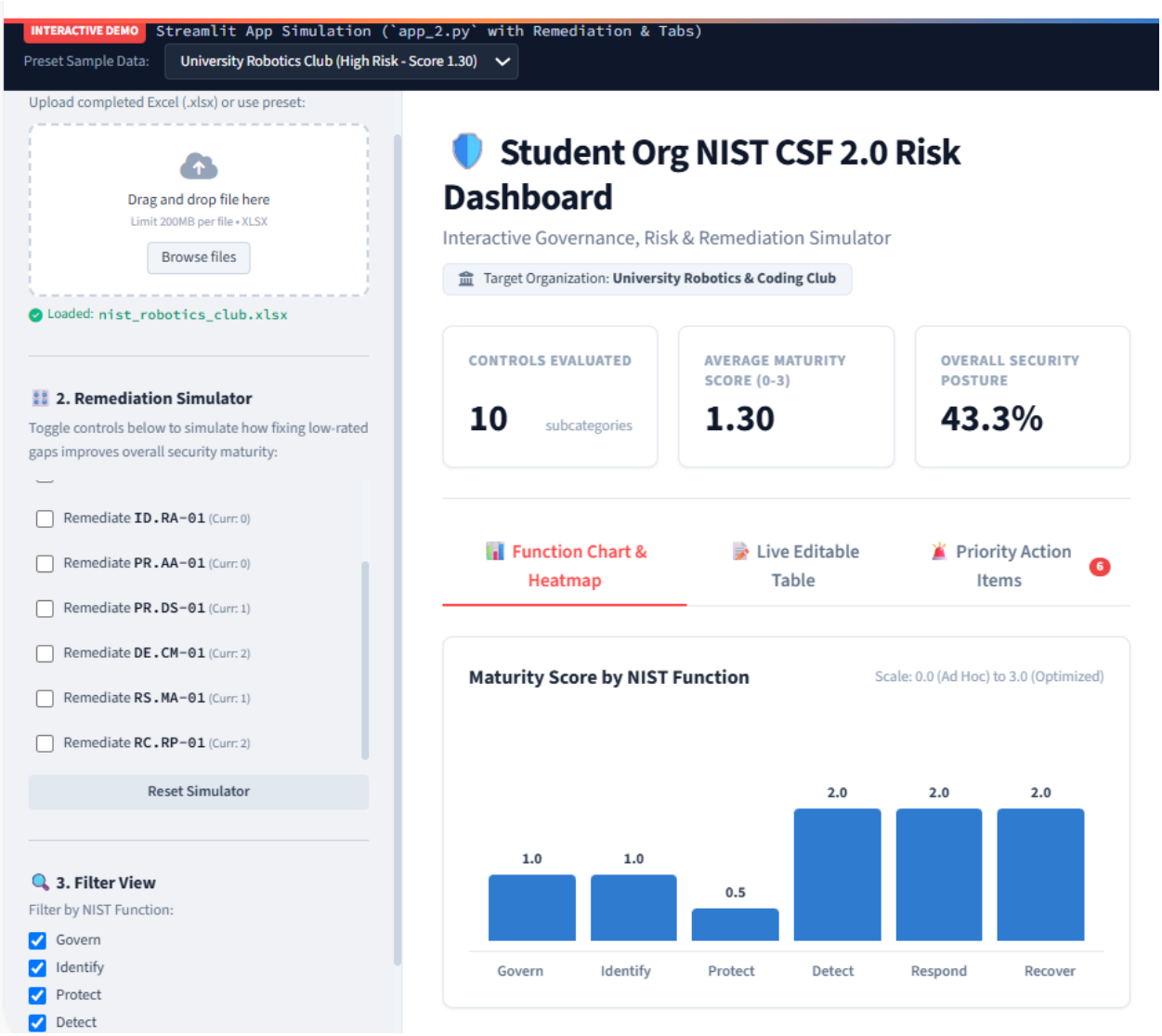
Data Visualizations

- Function Maturity Bar Chart (st.bar_chart): Automatically groups the assessed controls by their NIST function (Govern, Identify, Protect, etc.) and renders a visual bar chart displaying average scores.
- Color-Coded Heatmap Table: Automatically applies background highlights to the rating cells in the data table using soft colors ($0 =$ Red, $1 =$ Orange, $2 =$ Yellow, $3 =$ Green).
- Dynamic Metric Cards (st.metric): Displays top-level KPI cards with live calculations for total controls evaluated, average maturity score, and overall security posture percentage.

Interactive Features

- Drag-and-Drop File Uploader (st.sidebar.file_uploader): Allows students to upload any completed .xlsx assessment sheet directly into the web app through the sidebar without needing local folder placement.
- CSV Export Button (st.download_button): Lets users instantly download their modified or simulated assessment data as a CSV file with one click.
- Live In-App Data Editor (st.data_editor): Allows users to double-click and directly edit ratings (0–3) or update audit notes right inside the table during a presentation
- "What-If" Remediation Simulator (st.sidebar.checkbox): Automatically detects controls rated below Level 3 and generates interactive checkboxes in the sidebar. When toggled, it simulates fixing those gaps in real time, instantly updating the score metrics and showing delta changes (e.g., +0.45 score increase).
- Interactive Expandable Alert Cards (st.expander): Automatically extracts critical or high-risk gaps (rated 0 or 1) and displays them as expandable warning cards complete with specific evidence and audit notes
- Tabbed Interface (st.tabs): Organizes the page into three clean, clickable tabs:
    - Function Chart & Heatmap
    - Live Editable Table
    - Priority Action Items

Mockup



Grading Rubric for Final Project

Item	Grade (% of project grade)
Milestone 1	5
Milestone 2	5
Excel tool functionality (Proper dropdowns, correct roll-up formulas, automated maturity percentage, and effective conditional formatting heatmaps)	15
Governance and evidence logic (Depth and accuracy of evidence notes justifying why each control received a rating of 0, 1, 2, or 3 based on real operations.)	15
Two interactive features	20
One visualization	10
Presentation	30
Total	100

By the end of today's class, your team should have completed Milestone 1. This milestone is worth 5% of your project grade.

a. Completed and submitted rules of engagement on Canvas
b. Selected your student org
c. Planned your two interactive features and one visualization
d. Built the Excel workbook.
e. Submitted (b) and (c) in a one-page proposal on Canvas
f. Uploaded Excel workbook on Canvas (with blank cells – that is fine for today!)
g. Create Github and Streamlit accounts for every member of the team.

Final Project Deliverables

1. Excel analysis: The Excel dataset with its four tabs (submit on Canvas)
2. Streamlit dashboard: Include at least one visualization and two interactive features.
3. Risk exposure: Top 3 operational risk exposures identified (e.g., *offboarding gap, unencrypted payment data on personal drives*)
4. Prioritized Remediation Roadmap (actionable, zero-cost fixes for the student org, such as adopting university cloud storage or setting up an offboarding checklist)
5. Limitations: Explain the limitations of your study.
6. Presentation: Demonstrate the dashboard, present (3), (4), and (5) above, submit slide deck on Canvas.

Note: The project is designed to provide real-world skills and to nudge you out of your comfort zones. If done well, your group will end up with a real-world artifact that you can proudly display on your resumes and talk about in interviews. You will learn how to incorporate AI into your workflows, work with GitHub and Streamlit, do some rudimentary coding applications, and demonstrate risk management for a real-world scenario.

There are no separate points for peer review of teammates. However, you are encouraged to contact me directly if you face problems regarding teammates who do not participate or contribute to the project. I will treat all concerns confidentially. If you have concerns to report to me, please do so (in email or in person) no later than December 8, 2026. Any concerns reported after that date will not be considered. If more than two students approach me with concerns about a specific individual on their group who is not doing their part, that individual automatically stands to lose 10% of their project grade. Also, do not turn this into a game of revenge and payback. The solution to good teamwork is to start early, communicate often, and be mindful of everyone's circumstances and the greater good. Always begin with: how can I help?